UNIVERSITAT POLITÈCNICA DE CATALUNYA
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FIB

# Autonomous Navigation PROJECT




Robotics

Bachelor's Degree in Informatics Engineering

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# Autonomous Navigation & Multi-Stage Mission Execution

## 1. Project Objective

The goal of this project is to design and implement an autonomous software system that allows a TurtleBot3 Burger to complete a mission. You must integrate kinematics, sensor perception, obstacle avoidance, localization, and path planning to solve real-world navigation challenges in a dynamic environment.

## 2. The Mission

The mission is divided into three sequential phases. The robot must evolve without external assistance throughout all stages.

Phase I: Global Navigation & Obstacle Avoidance

Starting at an assigned initial pose (x_i, y_i, theta_i) within Zone 1, the robot has to navigate to a destination pose (x_f, y_f, theta_f) in Zone 2, ensuring the path passes through Waypoints E, Door and Q.

The environment may contain static or dynamic obstacles not present in the initial map. You have to implement a real-time obstacle detection and avoidance procedure.

Phase II: Perception & Area Exploration

Once the robot reaches the final pose (within Zone 2), it has to explore the "Corridor" area to locate a specific charging station. The station is a 40 cm x 40 cm square structure supported by four pillars (approx 5 cm diameter).

Once the Coridor area is completely explored and the station is identified, the robot has to return to "Punt Base". The “corridor” area measures no more than 12 by 12 meters.

While completing Phase I and Phase II the robot has to build his own environment map.

You must submit the map file (.yaml and .pgm) generated by the robot during the mission. This map must be the result of the robot’s own sensors and SLAM implementation processed during the completion of the first two phases, demonstrating the robot's ability to reconstruct the environment accurately (walls, other fixed objects not included in the prior map, obstacles, …, and the charging station pillars).

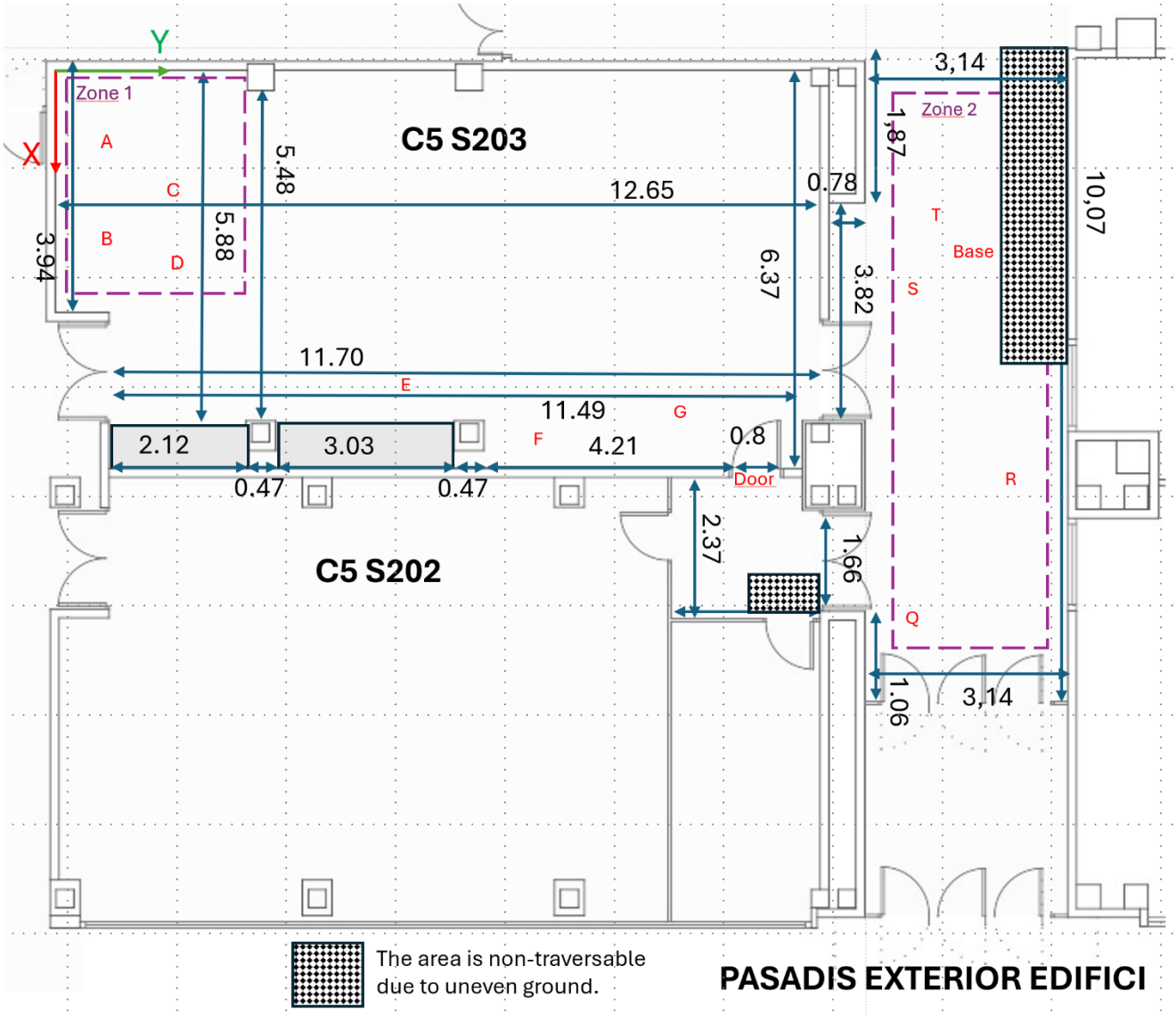
Phase III: Precision Docking

Finally, from "Punt Base," the robot has to execute a high-precision maneuver to park the robot exactly in the middle of the charging station. This action completes the robot mission.

The navigation system must prioritize the physical integrity of the TurtleBot3 platform. Path planning must be intrinsically safe, incorporating a conservative clearance buffer from all obstacles. In alignment with professional robotics standards, any physical collision with the environment will result in a severe grade penalty.

## 3. Environment & Coordinate Data

You can use the provided environment map to inform your global path planner. All coordinates are in meters relative to the map origin. Use of the map is no mandatory.



Key Reference Points

| Point | Coordinates (x, y) | Point | Coordinates (x, y) |
|---|---|---|---|
| Punt A | (0.7, 0.92) | P Base | (3.340 15.700) |
| Punt B | (3.102, 1.722) | Punt Q | (8.970, 14.497) |
| Punt C | (2.078, 2.517) | Punt R | (7.500, 16.170) |
| Punt D | (5.285, 2,517) | Punt S | (3.540, 14.497) |
| Punt E | (5.082, 6.122) | Punt S | (3.675, 14.190) |
| Punt F | (5.679, 8.528) | Punt T | (2.136, 15.300) |
| Punt G | (5.482, 10.930) | | |
| Door | (6.490, 11.682) | | |

Warning: The cell (0, 0) of the maps available at Atenea does not match the world reference system.

## 4. Evaluation Criteria

The project will be presented in a live demo at the lab, using the real TB3 robot.

Your project will be graded based on the following metrics:

- Engineering Standards: Code quality, project methodology, and technical documentation.
- Navigation Quality: Safety of the path, total distance traveled, and time efficiency.
- Localization Accuracy: Precision of the final pose, waypoints and in the charging station.
- Perception & Avoidance: Effectiveness of the sensor analysis in detecting pillars and avoiding unknown obstacles.

Remember: Zero-tolerance for collisions. Any impact between the robot and its surroundings—static or dynamic—will be heavily penalized during the final assessment.

## 5. Deliverables

Teams must submit a compressed file containing:

- Declaration of the use of generative AI.
- Package / Source Code: A complete ROS 2 package.
- Technical Report: System Architecture Diagram: A simple block diagram showing the ROS2 topics and nodes used. Algorithm Description: Brief explanation of the logic used for Phase I and Phase II. Obstacle Avoidance Strategy: Description of how the LiDAR data is used to override the predefined path.
- Map of the environment generated by the robot when completing Phase I and II.
- Mission Log (CSV/TXT)
- A log file generated by the robot during the run containing:
    - timestamp
    - current_phase (I, II, or III)
    - robot pose (robot_x, robot_y, robot_yaw)
    - Position of obstacles detected within the passadis area and charging station position (defined as his pillar position)
- Video Demonstration: A short clip of the robot completing the full mission.